Mo Malekpour

Canada, Montreal (open to relocate) | momalekpour98@gmail.com | +14383656290
malekpour.io | linkedin.com/in/momalekpour | github.com/momalekpour | google scholar

Summary

AI Software Engineer focused on data systems, backend architecture, and practical LLM applications. I build systems at the intersection of AI and databases, including Text-to-SQL with LLM-router and novel evaluation framework published at NeurIPS and ICDE. Passionate about AI backend/infra and turning research ideas into robust, scalable, production-ready systems.

Education

MASc in Computer Engineering @ *Polytechnique Montreal* 08/2023 – 12/2025
- Nominated for the Best Master's Thesis Award, 2025
- Courses: Engineering AI Software Systems, Representation Learning, Advanced Concepts of Cloud Computing (GPA 4.0)
- TA: Files and Databases

BSc in Computer Science, minor in Economics @ *AmirKabir University of Technology* 09/2017 – 08/2022
- TA: Fundamentals of Programming, Advanced Programming, Principle of Software Design, Data Science, ML

Work Experience

Graduate Research Assistant @ *Mila - Quebec Artificial Intelligence Institute* 08/2023 – 12/2025
- Member of Data & AI Systems lab, focused on agentic AI and data systems, specifically Text-to-SQL, and exploring DB internals.
- Implemented VortoSQL (NeurIPS'24), an end-to-end Text-to-SQL system via LLM router, reducing pipeline costs by up to 40%.
- Developed SQLMorph (ICDE'26), a Text-to-SQL eval framework with novel fine-grained metrics and query mutation techniques.
- Implemented a wrapper on Text-to-SQL major benchmarks for evaluating in a holistic environment.

Data Engineer @ *Divar* 02/2022 – 11/2022
- Worked within the core Business Insight team for Divar's e-commerce platform with 30M+ monthly active users.
- Designed and implemented a company-wide KPI system as the single source of truth, with 50+ fine/coarse-grained metrics.
- Built and maintained 20+ ETL pipelines using Python, SQL, PySpark, Hadoop HDFS, Apache Airflow and Metabase.
- Delivered many ad-hoc analyses to uncover inefficiencies, identify trends, and produce dashboards and predictive models.

Software Engineer Intern @ *Segmentino* 06/2020 – 09/2020
- Worked at Segmentino, an AI-powered marketing automation platform enabling omnichannel engagement and retention.
- Developed `mslib`, an internal Python SDK for Node.js microservices, reducing integration time and development overhead.
- Built `enlib`, a backend engine for the marketing journey designer, transforming user-created workflows into dynamic DAGs.

Software Engineer Intern @ *Institute for Research in Fundamental Sciences (IPM)* 06/2019 – 09/2019
- Improved data infrastructure for monitoring systems in Technical Development Division of the National Observatory.
- Redesigned the MySQL weather monitoring database and built Flask APIs to streamline environmental data for R&D teams.
- Built a web server for Robo-DIMM (night-sky telescope) using socket-based communication and modular system design.

Technical Skills

Programming & Scripting: Python, C++, SQL, Shell, Object-Oriented Programming (OOP)
Data Management & Database: PostgreSQL, MySQL, DuckDB, Trino, Apache Airflow, Apache Spark (PySpark), Apache Flink, Apache Kafka, Apache Iceberg, Parquet, HDFS, ETL/ELT, Stream Processing, Data Modeling, Data Warehousing
Machine Learning & AI: NumPy, Pandas, Scikit-Learn, PyTorch, Hugging Face (transformers), LangChain, LangGraph, LlamaIndex, OpenAI API, Model Fine-tuning, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Agentic AI workflows, MLOps
Backend Development: Flask, FastAPI, RESTful APIs, Networking (HTTP, UDP, TCP/IP, TLS)
Cloud & DevOps: Git, Docker, AWS (EC2, S3, Lambda, CloudWatch), CI/CD (GitHub Actions), Unix/Linux
Business Intelligence: Matplotlib, Seaborn, Plotly, Streamlit, Metabase, Data Visualization

Publications & Projects

SQLMorph: Query Mutation and Fine-Grained Metrics for Text-to-SQL Evaluation @ *ICDE 2026* 02/2026
- M Malekpour, M Riahi, M Lamothe, A Mhedhbi

Towards Optimizing Text-to-SQL Pipelines via LLM Routing @ *NeurIPS 2024 TRL* 11/2024
- M Malekpour, N Shaheen, F Khomh, A Mhedhbi

Machine Learning Predictions for Optimal Cement Content in Sustainable Concrete @ *JOBE 82* 04/2024
- M Shahrokhishahraki, M Malekpour, S Mirvalad, G Faraone

DailyDiet: Full-Stack Nutrition Management Platform @ *GitHub* 06/2020
- Python, Flask, PostgreSQL, Elasticsearch, Heroku